A synthetic magnetic-signature dataset with electrochemical degradation labels for lithium-ion batteries

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Abstract

Battery diagnostics is commonly based on terminal electrochemical measurements, while magnetic sensing can provide complementary spatial information about internal current distribution and magnetisation. Public datasets combining magnetic measurements with degradation labels remain scarce. We present MagBridge-Battery v1.0, a reproducible synthetic benchmark built from two independent public sources: battery magnetometry provides measured magnetic morphology, while PulseBat provides state-of-health, state-of-charge, and pulse-derived features from retired lithium-iron-phosphate cells. Because the sources contain different cells and chemistries and are not experimentally paired, the generated signatures should be interpreted as synthetic benchmark data rather than measured magnetic ageing trajectories. The release contains 6,760 samples: 5,600 conditioned signatures, 600 controlled sensor-anomaly samples, and 560 low-voltage extrapolation-regime samples. Each sample contains six length-100 magnetic channels with provenance metadata. A cell-disjoint primary split prevents conditioning-cell overlap and keeps synthetic anomalies with their clean parents. Reference tasks assess recovery of the imposed state-of-health conditioning and its derived second-life class, plus anomaly detection and anomaly-subtype classification. The dataset is intended for reproducible benchmarking and method development, not safety-critical battery diagnosis.

Background & Summary

Data-driven battery health diagnostics rely primarily on voltage, current, temperature, and impedance measurements. Most public battery datasets reflect this measurement modality. Widely used resources such as the Stanford/MIT cycling dataset [1] and the PulseBat retired-cell pulse-test dataset [2, 3] provide electrochemical measurements across a range of degradation states, with state-of-health (SOH) information and, in some cases, longitudinal degradation trajectories. These resources have supported research on cycle-life prediction, capacity-fade estimation, state estimation, and second-life battery assessment. Recent experimental studies have also examined SOH estimation specifically for second-life batteries using partial operating profiles and electrochemical-impedance measurements [4, 5].

Recent advances in lithium-ion battery-state estimation include hybrid deep-learning strategies for state-of-charge (SOC) estimation in large-scale energy-storage systems [6], history-aware recurrent models for joint SOC and state-of-energy (SOE) estimation [7], and Bayesian-optimised bidirectional recurrent models combined with filtering for high-precision SOC estimation [8]. These studies demonstrate progress in temporal modelling and hybrid estimation using electrochemical and operational measurements. However, they do not address the separate scarcity of magnetic measurements paired with degradation labels. MagBridge-Battery addresses this complementary data gap through a reproducible synthetic benchmark, without claiming experimentally validated magnetic ageing behaviour.

Magnetic sensing provides a complementary measurement modality because magnetic fields around a cell depend on internal current distributions and magnetic responses that are not directly observable from terminal measurements alone. Previous studies have demonstrated magnetic characterisation of lithium-ion cells using several sensing approaches, including magnetometry and magnetic imaging [9, 10, 11, 12, 13, 14]. Recent work has further demonstrated magnetic imaging of current density in lithium-ion batteries [14] and operando magnetic microscopy of battery dynamics [13]. Applied initiatives such as the QuaLiProM project are also investigating atomic magnetometry combined with AI for SOH-oriented classification of aged lithium-ion cells and second-life assessment [15]. These studies establish magnetic sensing as a potentially informative complementary modality, but they do not provide a large public dataset in which magnetic measurements are experimentally paired with degradation labels across many aged cells.

Public resources linking these two types of information therefore remain limited. The Open Science Framework (OSF) archive associated with Mohammadi, Ilott, and Jerschow [11, 16] provides experimentally measured magnetic scans of a lithium-ion cell at five operating voltages, but does not provide SOH-labelled ageing trajectories across multiple cells. PulseBat [2, 3], in contrast, provides pulse-test measurements and degradation-related information from retired lithium-ion cells but contains no magnetic measurements. Importantly, these sources represent different cells, experimental settings, and cell chemistries and are not experimentally paired.

MagBridge-Battery was constructed as a reproducible cross-source synthetic benchmark using these complementary resources. Measured OSF magnetic data provide the magnetic-morphology scaffold. PulseBat supplies the SOH and SOC conditioning variables, while its pulse-voltage features are used upstream to derive the operating voltage.

The bridge uses operating voltage, SOH, and SOC to generate controlled synthetic variations of the magnetic scaffold. No OSF magnetic measurement is assumed to correspond to an individual PulseBat cell, and the imposed SOH dependence is not an experimentally established magnetic ageing mechanism. The resulting signatures are therefore synthetic benchmark instances, not measured magnetic degradation trajectories.

The release contains 6,760 samples: 5,600 conditioned magnetic signatures, 600 controlled sensor-anomaly samples, and 560 low-voltage extrapolation-regime (Regime-B) samples. Each sample contains six length-100 magnetic channels with provenance metadata, and a cell-disjoint primary benchmark split is provided. The dataset is intended for reproducible method development, sensitivity analysis, and comparison of algorithms under a documented synthetic construction. Reference tasks are provided for SOH regression and second-life classification, which assess recovery of the imposed SOH conditioning rather than detection of physical magnetic ageing, together with anomaly detection and anomaly-subtype classification. MagBridge-Battery is not intended as a substitute for experimentally paired magnetic–electrochemical measurements or for direct battery-health or safety-critical decision making. Instead, the bridge assumptions, generation parameters, and provenance are exposed so that methods developed on the benchmark can be evaluated against future paired measurements when such data become available.

Methods

Source data

MagBridge-Battery draws on two independent public data sources that provide complementary information and are not experimentally paired.

Magnetic-morphology source. Magnetic morphology is derived from the OSF battery magnetometry archive of Mohammadi, Ilott, and Jerschow [11, 16] (OSF DOI: `10.17605/OSF.IO/CW8ZV`). The archive contains magnetic-field measurements from a single lithium-ion pouch cell at five operating-voltage anchors $\{2.54, 2.81, 3.00, 3.10, 3.34\}$ V, with 41 scan posi-

tions at each voltage. Here, a scan position denotes one of the 41 spatial measurement locations at which the magnetic field was sampled around the cell for a given voltage anchor. The source experiment used a Powerstream PGEBNMU53040 pouch cell with approximate dimensions $2.9 \times 3.9 \times 0.7$ cm and a rated capacity of 600 mAh. The cell comprises a graphite anode and a nickel–manganese–cobalt-oxide (NMC) cathode [11]. The magnetic field was measured using atomic magnetometers while the cell was spatially scanned within a magnetically shielded susceptometry setup. These measurements provide the experimentally observed signal-domain morphology used to construct the bridge. The archive does not provide SOH-labelled ageing trajectories across multiple cells; consequently, no SOH value in MagBridge-Battery is interpreted as an experimentally measured health state of the OSF cell. The OSF source is treated only as the experimentally measured magnetic-morphology source; no chemistry match, ageing-history match, or cell-level correspondence with the PulseBat conditioning population is assumed. The OSF archive does not specify an explicit licence. Prof. Alexej Jerschow, an author of the archive, confirmed by email that the data may be reused with citation; this correspondence is available to the editors upon request.

Electrochemical-conditioning source. SOH, state-of-charge (SOC), and 21-dimensional pulse-voltage features are drawn from PulseBat [2, 3], specifically version v3 of the TBSI-Lijing Battery Dataset archived on Zenodo (DOI: `10.5281/zenodo.13360631`), a public pulse-test dataset of retired lithium-ion cells with heterogeneous retirement conditions. PulseBat uses constant-current–constant-voltage (CCCV) capacity-calibration results as the SOH benchmarking capacity values, while pulse-response measurements are used to derive the 21-dimensional feature representation.

MagBridge-Battery uses the calibrated SOH values supplied by PulseBat directly as conditioning labels. In the upstream PulseBat processing pipeline, SOH is defined as

$$\mathrm{SOH} = \frac{Q}{Q_n}, \tag{1}$$

where Q is the calibrated discharge capacity obtained from the capacity calibration test and Q_n is the nominal battery capacity. For the LFP subset used in MagBridge-Battery, $Q_n = 35$ Ah.

PulseBat determines the calibrated capacity using constant-current–constant-voltage (CCCV) charging followed by constant-current discharge. The cell is charged at 1C to the upper cut-off voltage and held at that voltage until the current decreases to 0.05C. It is then discharged at 1C to the lower cut-off voltage. For the 35-Ah LFP cells, the upper and lower cut-off voltages are 3.65 V and 2.50 V, respectively.

The resulting discharge capacity is treated as the calibrated battery capacity. MagBridge-Battery does not recompute SOH from the 21-dimensional pulse features; it uses the upstream PulseBat SOH values directly.

The pulse tests were conducted across SOC levels from 5% to 50% under a controlled ambient temperature of 25 °C. MagBridge-Battery uses exclusively the 35-Ah lithium-iron-phosphate (LFP) subset of PulseBat. The PulseBat records supply the conditioning-cell identifiers, calibrated SOH values, SOC values, and pulse-voltage features used upstream to derive the operating voltage and retained in the released metadata; they do not contain magnetic measurements. The associated PulseBat project repository states that the raw data and processed features are available under the MIT licence.

The two sources are used with explicitly separated provenance: OSF provides the measured magnetic-morphology scaffold, whereas PulseBat provides the independent electrochemical conditioning population. No cell-level pairing, chemistry matching, ageing-history correspondence, or transfer of magnetic ageing behaviour between the source populations is assumed. In particular, the OSF NMC morphology is used only as a measured signal-domain scaffold; the LFP conditioning variables are applied as synthetic benchmark controls, and no chemistry-specific NMC-to-LFP ageing relationship is assumed. Consequently, the SOH-conditioned signatures

are synthetic benchmark constructions rather than experimentally measured magnetic ageing trajectories.

Table 1: Comparison of the two independent upstream data sources used in MagBridge-Battery. The sources are not experimentally paired; the OSF dataset provides magnetic morphology, whereas PulseBat provides independent electrochemical conditioning variables.

Characteristic	**OSF magnetometry**	**PulseBat**
Role in MagBridge-Battery	Magnetic-morphology source	Electrochemical-conditioning source
Population used	Single pouch cell	Retired 35-Ah LFP cells
Cell chemistry	Graphite anode / NMC cathode	LFP
Rated capacity	600 mAh	35 Ah
Measurements used	Magnetic-field scans	CCCV capacity calibration and pulse-voltage measurements
Operating conditions	Five voltage anchors: 2.54, 2.81, 3.00, 3.10, and 3.34 V	Pulse tests at 5–50% SOC and 25 °C
Generation inputs	Magnetic morphology	SOH, SOC, and pulse-voltage features used to derive operating voltage
SOH information	No	Yes, calibrated SOH values
Magnetic measurements	Yes	No
Relationship between sources	Independent cells with different chemistries; no cell-level pairing or correspondence in ageing history between the OSF magnetic measurements and PulseBat conditioning records is assumed.	

Bridge overview

The bridge is a deterministic function for fixed inputs, configuration, and random seed: $\mathcal{B}(v, \mathrm{SOC}, \mathrm{SOH}; \theta, s) \to \mathbf{X} \in \mathbb{R}^{T \times C}$, where v is the operating voltage, θ denotes the bridge configuration, and s is the per-sample random seed. The generated magnetic signature has length $T = 100$ across $C = 6$ channels.

For the released grounded dataset, the operating voltage v is derived upstream from the mean of the 21 PulseBat pulse-voltage features and clipped to the grounded range $[3.06, 3.34]$ V. The pulse-feature vector is retained in the released provenance metadata but is not used directly by the bridge after the operating voltage has been determined. The OSF measurements and PulseBat-LFP records enter the bridge as independent sources with distinct roles: the former provides the magnetic-morphology scaffold, whereas the latter provides electrochemical conditioning variables. No sample-level pairing between the two sources is assumed.

In practical terms, generation proceeds in four steps. First, the requested voltage selects or interpolates an OSF-derived magnetic morphology. Second, PulseBat SOH is used to impose controlled synthetic SOH-dependent perturbations on that morphology, while SOC controls the low-frequency fluctuation term; the operating voltage is derived upstream from the PulseBat pulse-voltage features. Third, the perturbed representation is decoded back to the six-channel signal domain. Finally, sensor and SOC-dependent noise are added to produce the released synthetic signature.

The configuration θ collects all tunable generation parameters and is fixed at bridge instantiation. Generation is reproducible: a given (request, θ, seed) tuple always produces the same output, and the configuration is recorded by a hash in the released manifest. Figure 2 illustrates how the OSF-derived morphology bank, regime-dependent SOH conditioning, decoding operations, and noise generation contribute to the construction of grounded and Regime-B samples.

The resulting release is organised into three sample groups, as summarised in Fig. 1. Of

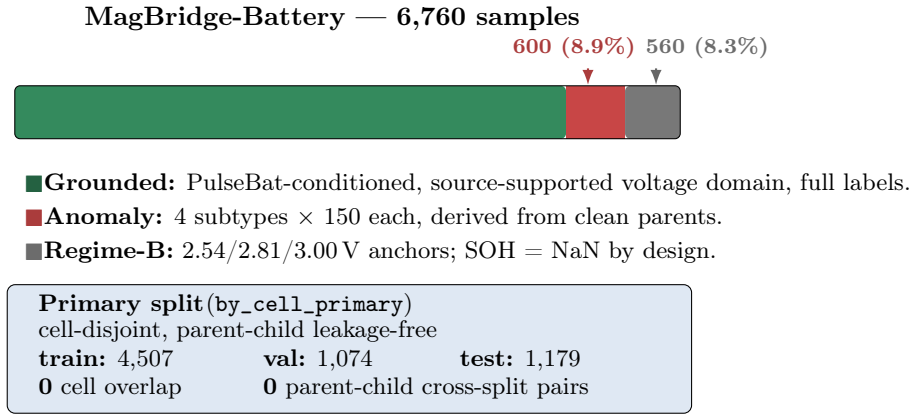


Figure 1: Composition and primary split of MagBridge-Battery. The dataset contains 6,760 samples: 5,600 grounded samples constructed from OSF-derived magnetic morphology conditioned using PulseBat-LFP conditioning variables, 600 synthetic anomaly samples derived from clean parent samples, and 560 low-voltage Regime-B samples for which SOH is undefined by design. Bar width is proportional to sample count. The primary benchmark split, `by_cell_primary`, is cell-disjoint and parent-child leakage-free. Here, "grounded" denotes construction within the source-supported operating and conditioning domains and does not imply experimentally paired magnetic-field/SOH measurements.

the 6,760 samples, 5,600 are grounded samples constructed within the source-supported operating domain using PulseBat-LFP conditioning variables, 600 are controlled synthetic anomaly samples derived from clean parents, and 560 are low-voltage Regime-B samples for which SOH is undefined by design. The primary `by_cell_primary` split is cell-disjoint and keeps each synthetic anomaly sample in the same partition as its clean parent.

Regime classifier

The bridge operates in two voltage regimes, both recorded in the `regime` metadata field. In the *grounded* regime (`regime=grounded`, $v \in [3.06, 3.34]$ V), PulseBat electrochemical conditioning is available. For each request, the base magnetic morphology is constructed from the OSF voltage anchors that bracket the requested voltage according to the interpolation procedure described below; the 3.06 V value defines the conditioning-regime boundary and is not itself an OSF measurement anchor. The PulseBat-derived SOH and SOC variables are then applied as synthetic conditioning variables.

In the *extrapolation* regime (`regime=extrapolation`, also referred to as Regime-B; $v \in [2.54, 3.06)$ V), PulseBat SOH conditioning is not applied. Samples in this regime are generated from the OSF morphology bank and are intended for low-voltage and out-of-distribution evaluation rather than SOH regression. Voltages outside $[2.54, 3.34]$ V are rejected. Every generated sample records its regime in the metadata.

Here, "extrapolation" refers to operation outside the PulseBat-conditioned grounded regime; the Regime-B voltages themselves correspond to measured OSF voltage anchors and therefore do not represent extrapolation beyond the OSF magnetic measurement range.

Among the OSF voltage anchors, 3.10 V and 3.34 V lie within the grounded regime and therefore provide direct anchor points for the distributional validation reported in Technical Validation.

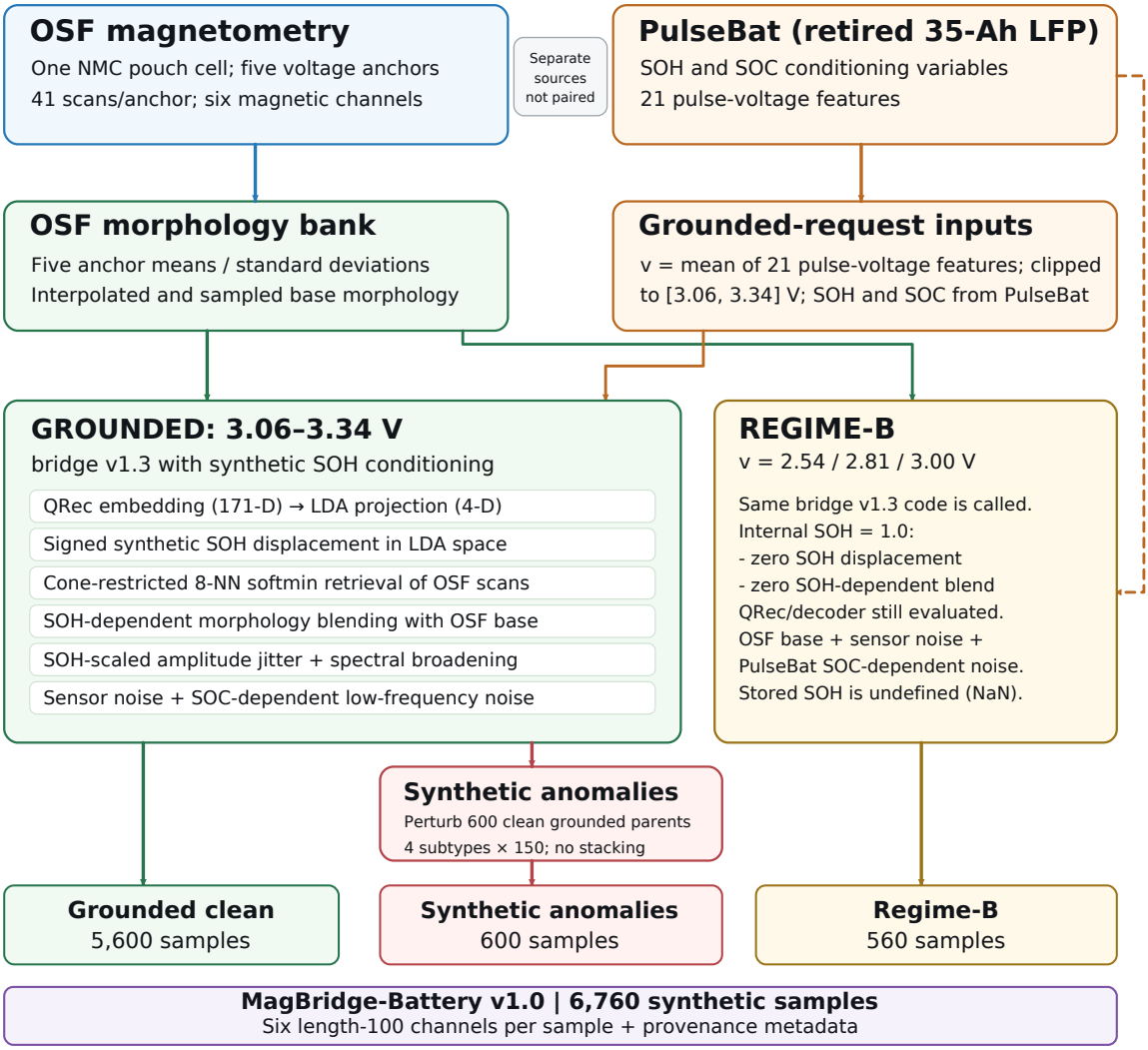


Figure 2: Architecture of the MagBridge-Battery synthetic data-generation pipeline. OSF magnetometry (blue) supplies the measured magnetic-morphology scaffold, while PulseBat (orange) provides independent electrochemical conditioning variables. For grounded samples, the bridge combines voltage-dependent morphology with synthetic SOH conditioning through a fixed quantum-reservoir embedding, LDA projection, cone-restricted decoding, morphology blending, and noise generation. Synthetic sensor anomalies are derived from clean grounded samples. Regime-B samples are generated at low OSF voltage anchors without SOH-dependent conditioning; SOC-dependent noise remains active. The release contains 5,600 clean grounded samples, 600 synthetic anomalies, and 560 Regime-B samples. The two upstream sources are not experimentally paired, and no raw upstream files are redistributed.

Morphology bank

The OSF archive is canonicalised into per-anchor empirical statistics. At each of the five anchor voltages, the bridge extracts a mean trajectory $\boldsymbol{\mu}_v \in \mathbb{R}^{T\times C}$ and a per-timestep, per-channel standard deviation $\boldsymbol{\sigma}_v \in \mathbb{R}^{T\times C}$ over the 41 scan positions. For a generation request at voltage v bracketed by anchors $v_l \le v \le v_h$, the bank forms a base sample by linear interpolation,

$$\boldsymbol{\mu}(v) = (1-\alpha)\,\boldsymbol{\mu}_{v_l} + \alpha\,\boldsymbol{\mu}_{v_h}, \qquad \alpha = \frac{v - v_l}{v_h - v_l}, \tag{2}$$

$$\boldsymbol{\sigma}(v) = \sqrt{(1-\alpha)\,\boldsymbol{\sigma}_{v_l}^2 + \alpha\,\boldsymbol{\sigma}_{v_h}^2}, \tag{3}$$

$$\mathbf{X}_{\text{base}} = \boldsymbol{\mu}(v) + \boldsymbol{\sigma}(v) \odot \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, I). \tag{4}$$

The measured OSF data therefore determine the anchor morphology, while intermediate-voltage and stochastic realisations are generated by the interpolation and sampling procedure above. At an exact anchor with no perturbation, the deterministic anchor-replica configuration is defined to return the empirical mean trajectory; this property is tested in Technical Validation.

SOH-conditioning module

The SOH-conditioning module is a synthetic component of the bridge that applies controlled SOH-dependent perturbations to the base morphology through a fixed feature representation, termed MagBridge-Embed in the released pipeline. Here, SOH conditioning refers to controlled synthetic conditioning using the PulseBat SOH variable and does not imply that the imposed perturbation represents an experimentally established magnetic ageing mechanism.

MagBridge-Embed is a 171-dimensional reservoir-derived embedding used as a reproducible latent space for defining and decoding morphology perturbations. In this release, the embedding is produced by a fixed quantum reservoir encoder following the reservoir-computing principle of Fujii and Nakajima [17]. The encoder uses 10 qubits arranged as 4 memory qubits and 6 processor qubits, two reservoir layers, and pooling over last, mean, and standard-deviation statistics. The reservoir circuit was implemented in PennyLane using the `default.qubit` simulator; the generation environment specified PennyLane version 0.40 or later. The reservoir parameters are not trained; only the downstream linear projections used inside the bridge are fitted. The encoder therefore acts as a deterministic feature transform within the released generation pipeline rather than as a trained quantum model. It is retained in MagBridge-Battery because it is the representation used to generate the frozen released artifact; its use does not imply a claim of quantum advantage or performance optimality. The dependence of benchmark utility on this representation is examined using matched alternative representations in Technical Validation.

The module first computes the embedding $\mathbf{e}_{\text{base}}$ of the base sample and projects it into a four-dimensional Linear Discriminant Analysis (LDA) subspace fitted on real OSF samples with the voltage anchors as classes,

$$\mathbf{z}_{\text{base}} = \big(\boldsymbol{\Sigma}^{-1/2}(\mathbf{e}_{\text{base}} - \boldsymbol{\mu}_e) - \bar{\mathbf{x}}\big)\mathbf{W}_{\text{LDA}}. \tag{5}$$

Here, μ_e and Σ denote the fitted embedding centring and scaling statistics used for standardisation, $\bar{x}$ is the LDA centring vector, and W_{LDA} is the fitted four-dimensional LDA projection matrix.

The fitted unit direction $\hat{\mathbf{d}}$ is aligned with the ordered OSF voltage-anchor centroids in the four-dimensional LDA space, with the positive direction corresponding to increasing anchor voltage. A synthetic SOH-conditioning displacement is then applied along the fitted direction $\hat{\mathbf{d}}$, with magnitude determined by $\delta = \max(0, 1 - \text{SOH})$,

$$\mathbf{z}_{\text{pert}} = \mathbf{z}_{\text{base}} + m\hat{\mathbf{d}}, \qquad m = -\gamma\delta. \tag{6}$$

This displacement is a benchmark-generation rule rather than an experimentally derived magnetic ageing trajectory. The fitted direction $\hat{\mathbf{d}}$ is not interpreted as unique, causal, or physically identified; alternative-direction ablations are evaluated in Technical Validation.

The perturbed representation is decoded back to the time domain by a cone-restricted k-nearest-neighbour softmin retrieval against the real OSF samples,

$$\mathbf{X}_{\text{decode}} = \sum_{i \in \mathcal{N}_k} w_i \mathbf{X}_i^{\text{OSF}}, \qquad w_i = \frac{\exp(-d_i/\tau)}{\sum_j \exp(-d_j/\tau)}, \tag{7}$$

with $k = 8$ neighbours and a 75° alignment cone. The decoded morphology is then blended with the base sample according to

$$\mathbf{X}_{\text{mod}} = (1 - \beta)\mathbf{X}_{\text{base}} + \beta\mathbf{X}_{\text{decode}}, \qquad \beta = \min(1, 1.5\delta). \tag{8}$$

In the released bridge implementation (v1.3), the SOH-dependent latent displacement is applied opposite to the fitted positive direction $\hat{\mathbf{d}}$, whereas the cone filter evaluates candidate alignment using the positive direction $\hat{\mathbf{d}}$. Specifically, candidate displacements are measured relative to $\mathbf{z}_{\text{base}}$, and their angular alignment is tested against $\hat{\mathbf{d}}$, rather than against the signed SOH-dependent displacement vector. Cone filtering is bypassed for $\delta < 0.02$, and the unrestricted candidate set is used when fewer than eight candidates satisfy the angular criterion. This convention is preserved in the frozen released dataset; a sign-consistent alternative is evaluated separately as a diagnostic in Technical Validation.

Two further SOH-scaled operations are applied after decoding. First, per-channel multiplicative amplitude jitter increases with $\delta = \max(0, 1 - \text{SOH})$. Second, zero-mean broadband Gaussian perturbations (referred to below as spectral broadening) are added with magnitude proportional to the same SOH difference. The released configuration sets the amplitude-strength and spectral-strength parameters to 0.30 and 0.05, respectively. These operations are part of the synthetic benchmark construction and are not calibrated representations of physical ageing. The PulseBat pulse-feature vector does not directly modulate these operations; in the released generation pipeline, it is used upstream to determine the operating voltage of each grounded request. The same configuration uses $\gamma = 800$, $k = 8$, $\tau = 50$, and a cone half-angle of 75°; the full configuration is pinned by hash in the manifest. Sensitivity to these generation parameters is evaluated in Technical Validation.

Noise model

Two additive noise sources are applied to the module output. Sensor noise is zero-mean Gaussian with per-channel standard deviation set to 5% of the standard deviation of the SOH-conditioned signal over its 100 time points for the corresponding channel. An SOC-dependent fluctuation adds a low-frequency component scaled by $|\text{SOC} - 50|/50$, implemented as boxcar-smoothed Gaussian noise normalised and scaled to 4% of the per-channel range. No additional SOH-dependent disorder term is applied, to avoid double-counting the spectral broadening already introduced by the SOH-conditioning module.

Synthetic sensor anomalies

For the 600 anomaly samples, a clean parent sample is generated by the pipeline above and one of four controlled perturbations is applied: `sensor_dropout`, `calibration_drift`, `temporal_warp`, or `periodic_interference` (150 samples each). These perturbations represent generic sensor and acquisition corruptions, including missing measurements, calibration drift, timing distortion, and periodic interference. Bias, drift, scaling, and dropout are established sensor-fault modes [18], while magnetic measurements are also susceptible to external interference that can require explicit compensation [19]. They are implemented as controlled benchmark stress tests rather than as experimentally calibrated fault distributions for a specific magnetometer or battery experiment. Their severity values should therefore be interpreted as reproducible generation settings, not as estimates of real-world fault prevalence or magnitude. The parent identifier is retained in `parent_sample_id`, and parent–child samples are kept within the same benchmark partition.

Reproducibility

Generation is fully deterministic given the configuration θ and per-sample seeds. The released manifest records the bridge version, a hash of the bridge hyperparameters, the schema version,

and per-sample generation seeds, allowing the released generation configuration to be identified and replayed. The bridge implementation, generation scripts, and unit tests are openly available (Code Availability).

Data Records

The MagBridge-Battery dataset is archived on Zenodo [20] under a Creative Commons Attribution 4.0 (CC-BY-4.0) licence. The release is distributed as a single bundle containing five Parquet data shards of 1,352 rows each, a metadata-only Parquet view, two benchmark-split files in JSON format, a generation manifest, a minimal Python loader, and SHA-256 checksums for every shipped file. The complete dataset contains 6,760 rows. Each row corresponds to one synthetically generated magnetic-signature sample and contains six length-100 magnetic channels, a shared length-100 normalised time grid, conditioning and provenance metadata, and anomaly-related fields where applicable.

Composition

The 6,760 samples are partitioned into three provenance groups. The grounded group contains 5,600 samples generated within the grounded voltage regime using PulseBat-LFP SOH and SOC as conditioning variables, with the operating voltage derived upstream from the PulseBat pulse-voltage features. These samples provide the clean labelled subset for SOH regression and second-life classification. The anomaly group contains 600 synthetic sensor-anomaly samples derived from clean grounded parents, with 150 samples each for `sensor_dropout`, `calibration_drift`, `temporal_warp`, and `periodic_interference`. Each anomaly row stores the identifier of its clean parent. The extrapolation-regime (Regime-B) group contains 560 low-voltage samples generated at the Regime-B anchors $\{2.54, 2.81, 3.00\}$ V. These samples lie outside the PulseBat-LFP conditioning range used in this release; therefore `soh`, `u_features`, and `second_life_class` are set to missing values by design. They are intended for low-voltage and out-of-distribution evaluation rather than SOH regression.

Signal channels

Each row contains six magnetic signal channels, each represented as a length-100 sequence: `B_s1Y`, `B_s1Z`, `B_s2Y`, `B_s2Z`, `B_s1C5`, and `B_s2C6`. The first four represent signed Y and Z field components for two sensors. The final two are signed source channels retained from the OSF archive. Although these channels are labelled as magnitude channels in the source archive, their values are signed and may be negative; they are therefore preserved as signed channels and should not be interpreted as strict $\sqrt{Y^2 + Z^2}$ magnitudes.

The `time_norm` column contains 100 evenly spaced values in $[0, 1]$. This reference grid is identical for every sample and therefore carries no sample-specific timing information. The `temporal_warp` anomaly perturbs signal values on this fixed grid rather than introducing an irregular per-sample timebase.

Metadata fields

Each row additionally carries identifier, provenance, conditioning-label, and anomaly-label fields. Table 2 documents these fields. SOH-related conditioning labels are populated for grounded and anomaly samples; for extrapolation-regime samples, they are `NaN` by design.

Benchmark splits

Two benchmark splits accompany the dataset. The primary split, `by_cell_primary`, is intended for all headline results. It is cell-disjoint and parent–child-leakage-free, with 4,507 training sam-

Table 2: Metadata fields in each MagBridge-Battery record. Chemistry, SOH, SOC, and pulse-feature fields refer to the PulseBat conditioning source and should not be interpreted as measured properties of the independent OSF magnetic-morphology source. Signal channels (`B_s1Y`, `B_s1Z`, `B_s2Y`, `B_s2Z`, `B_s1C5`, `B_s2C6`) and `time_norm` are length-100 numeric arrays and are not repeated here.

Field	Type	Description
`sample_id`	string	Unique identifier for the sample.
`parent_sample_id`	string	For anomalies, the clean parent's `sample_id`; else null.
`cell_id`	string	PulseBat conditioning-cell identifier for grounded and synthetic-anomaly samples; Regime-B samples use one of three voltage-specific pseudo-cell identifiers.
`generation_seed`	int	Random seed used to generate the sample.
`bridge_version`	string	Bridge version tag (`v1.3`).
`bridge_config_hash`	string	Hash of the bridge hyperparameters.
`schema_version`	string	Schema version tag.
`voltage`	float	Operating voltage in volts.
`soc`	float	PulseBat-derived state-of-charge conditioning value, percent.
`soh`	float	PulseBat-derived state-of-health conditioning value in $[0, 1]$; `NaN` for extrapolation-regime samples.
`chemistry`	string	Chemistry of the PulseBat conditioning cell (`LFP` in this release); this field does not describe the independent OSF magnetic-morphology source.
`regime`	string	`grounded` or `extrapolation`.
`nearest_anchor`	float	Nearest OSF voltage anchor, volts.
`u_features`	array	21-dimensional PulseBat-derived pulse-voltage vector used upstream to derive the operating voltage; `NaN` for extrapolation-regime samples.
`second_life_class`	string	`reuse` or `recondition`; `NaN` for extrapolation-regime samples.
`anomaly_flag`	bool	True for synthetic sensor anomalies and extrapolation-regime samples; the field therefore denotes departure from the clean grounded subset, not exclusively a sensor fault.
`anomaly_subtype`	string	One of four subtypes, or null.
`anomaly_origin`	string	Provenance tag for the anomaly perturbation.
`anomaly_severity`	float	Perturbation severity for anomalies.

ples, 1,074 validation samples, and 1,179 test samples. The split has zero PulseBat conditioning cells shared between partitions, zero clean-parent/anomaly-child pairs divided across partitions, and zero sample-identifier overlap.

The three Regime-B voltage-specific pseudo-cells are assigned one per partition under the deterministic split procedure: the training, validation, and test partitions contain 199, 178, and 183 Regime-B samples, respectively, ensuring that all three partitions retain Regime-B examples for the T3 benchmark.

A companion split, `by_record_optimistic_baseline`, is provided only as a leakage contrast. It contains quantified leakage, including 59 dataset cell identifiers appearing in more than one partition (56 PulseBat conditioning-cell identifiers and three Regime-B voltage-specific pseudo-cell identifiers) and 292 parent–child pairs split across partitions. This companion split should not be used for headline benchmarking; it is included to allow users to measure the inflation caused by record-level splitting.

Each split is stored as a JSON object containing sample-identifier arrays for training, validation, and testing, named `train_samples`, `val_samples`, and `test_samples`, respectively. The split files also include the corresponding conditioning-cell lists and split metadata.

Second-life label definition

The `second_life_class` field is a benchmark classification label derived from the PulseBat SOH conditioning value using a fixed threshold of SOH = 0.85. Samples with SOH > 0.85 are labelled `reuse`, whereas samples with SOH ≤ 0.85 are labelled `recondition`. This threshold is used to provide a reproducible binary benchmark task and should not be interpreted as an independently measured or universally applicable criterion for second-life suitability.

Data Overview

The released grounded-regime SOH values span [0.744, 0.962], reflecting the SOH range of the PulseBat conditioning records used in this release. Among the 5,600 grounded samples, 2,600 (46.4%) are labelled `reuse` and 3,000 (53.6%) are labelled `recondition`.

Technical Validation

Technical validation was performed through structural checks on every release candidate, distributional comparison against real OSF samples at the grounded voltage anchors, controlled ablations of the imposed SOH conditioning, and additional robustness analyses of representation choice, bridge hyperparameters, and generation randomness. Reference predictive models are also evaluated under the cell-disjoint benchmark protocol to assess whether the generated signatures support the intended benchmark tasks. These analyses are intended to support reproducible reuse, quantify sensitivity to design choices, and document the assumptions of the bridge. They do not establish physical equivalence to experimentally paired magnetic–SOH measurements, which are not provided by the source datasets used here.

Figure 3 summarises the validation and benchmark workflow used for the released dataset.

To provide reference points for reuse of the dataset, Table 3 reports performance on the four benchmark tasks using the cell-disjoint `by_cell_primary` split. Classical baselines are standard scikit-learn pipelines, including ridge regression, support-vector regression, random forests, logistic regression, ridge classifier, and linear support-vector classifier, while neural reference baselines comprise a multilayer perceptron, a one-dimensional convolutional network, and a long short-term memory network operating on the raw six-channel sequences. These results document the utility and difficulty of the released benchmark tasks rather than state-of-the-art predictive performance; more complex sequence architectures such as Transformers and temporal convolutional networks are outside the scope of this reference benchmark. Because the T2 second-life label is deterministically derived from SOH using the fixed 0.85 threshold, T2 should be interpreted as a benchmark classification view of the SOH conditioning rather than as independent evidence of a separately measured second-life property.

Neural models were trained using Adam with a learning rate of 10^{-3}, weight decay of 10^{-4}, and batch size 64 for at most 15 epochs. Inputs were standardised per channel using statistics computed from the corresponding training subsample. Training was stopped early when the training loss failed to improve by at least 10^{-4} for three consecutive epochs. The validation split was not used for model selection, and the SOH target was not standardised.

The stronger classical baseline performance should not be interpreted as evidence of an intrinsic advantage over neural models. The classical models operate on compact 57-dimensional summary descriptors comprising nine features per channel across the six signal channels, together with three cross-channel correlation features. In contrast, the neural baselines operate directly on the raw 6×100 sequences and were used as reference models rather than extensively tuned architectures.

The moderate training-set size relative to the raw sequence dimensionality may favour the regularised classical pipelines over the higher-capacity neural architectures evaluated here. To-

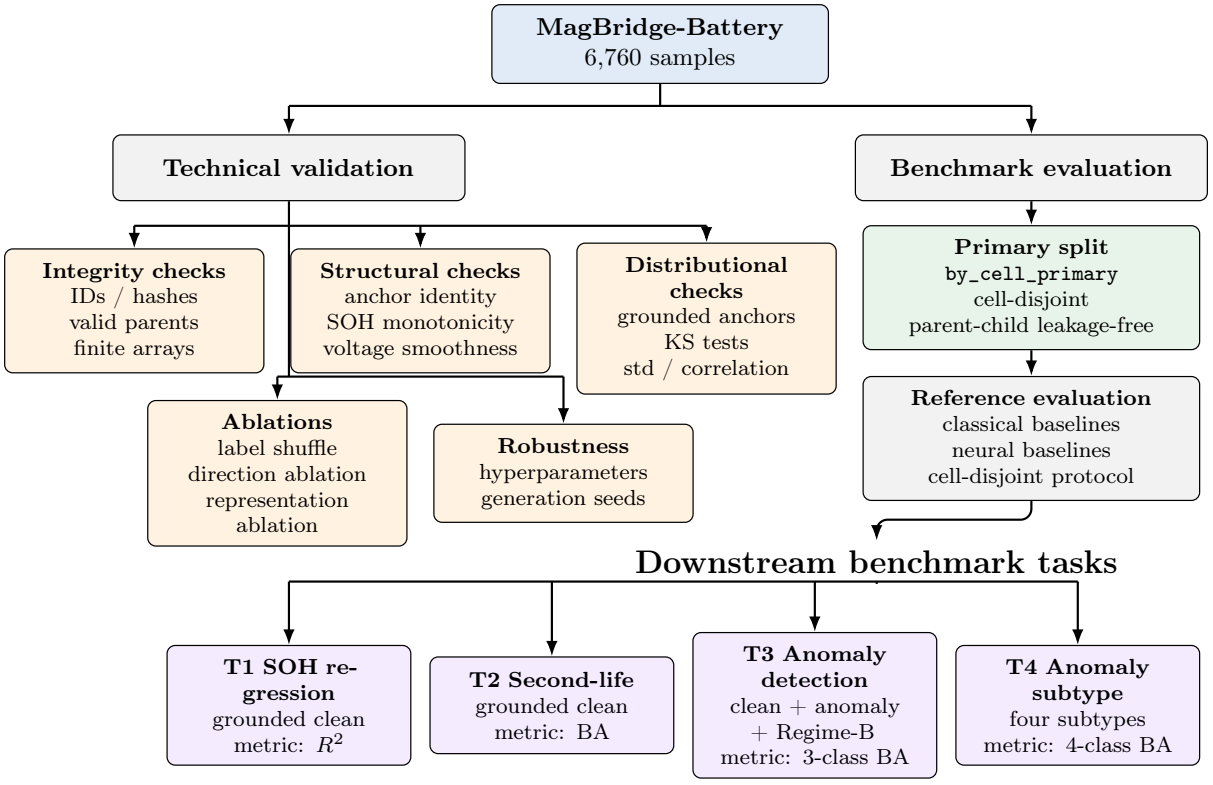


Figure 3: Technical validation and benchmark evaluation workflow for MagBridge-Battery. Validation includes structural, distributional, ablation, and robustness checks, while reference models are evaluated on four downstream tasks using the cell-disjoint `by_cell_primary` split. BA denotes balanced accuracy.

Table 3: Reference baseline performance on the `by_cell_primary` cell-disjoint split. T1 is reported using the coefficient of determination (R^2); T2–T4 are reported using balanced accuracy (BA). Classical results are mean ± standard deviation over five repeated random 80% subsamples of the training samples, while neural results are mean ± standard deviation over three repeated random 80% subsamples of the training samples. "Best classical" and "Best neural" denote the strongest reference model for each task under the released benchmark workflow. These values characterize the released benchmark tasks and are not intended as tuned performance ceilings.

Task	Metric	Best classical	Best neural
T1 SOH regression	R^2	0.675 ± 0.001	0.170 ± 0.240
T2 Second-life classification	BA	0.907 ± 0.001	0.858 ± 0.056
T3 Anomaly detection (3-class)	BA	0.789 ± 0.003	0.755 ± 0.028
T4 Anomaly subtype (4-class)	BA	0.725 ± 0.028	0.355 ± 0.024

gether with the limited tuning of the neural baselines, this provides a plausible explanation for the observed performance gap but is not established as its unique cause.

Accordingly, the comparison in Table 3 should be interpreted as descriptive reference benchmarking under the released workflows, rather than as a controlled comparison of the intrinsic capabilities of classical and neural model classes.

Matched-input classical–neural comparison

To investigate the influence of input representation on the observed classical–neural baseline differences, we performed an additional matched-input comparison on the frozen MagBridge-

Battery v1.0 release using the `by_cell_primary` split.

Four configurations were evaluated for each benchmark task: a classical Ridge or logistic-regression model and a multilayer perceptron (MLP), each trained using either the original 57-dimensional engineered descriptors or the flattened 600-dimensional raw magnetic signals. The same MLP architecture, comprising hidden layers of 64 and 32 units, was used for both input representations.

Five predetermined training-subsampling seeds were used. For each seed, all four configurations received the same 80% training-row subsample and were evaluated against identical held-out test samples. Model hyperparameters were selected using the archived validation partition. Four prespecified candidates were evaluated per configuration. The MLP used validation-based early stopping with a maximum of 80 training epochs. SOH targets were standardised using training-only statistics for MLP regression and transformed back before evaluation.

Table 4: Matched-input comparison on the released `by_cell_primary` split. Classical models are Ridge regression for T1 and logistic regression for T2–T4. Neural results use a fixed two-hidden-layer MLP. Values are mean ± standard deviation across five training subsamples. T1 uses R^2; T2–T4 use balanced accuracy. These results are separate from the original descriptive reference baselines.

Task	Classical 57D	MLP 57D	Classical 600D	MLP 600D
T1	0.675 ± 0.002	0.681 ± 0.004	-0.558 ± 0.038	-0.020 ± 0.017
T2	0.904 ± 0.003	0.912 ± 0.003	0.530 ± 0.013	0.574 ± 0.012
T3	0.723 ± 0.004	0.759 ± 0.066	0.681 ± 0.005	0.425 ± 0.133
T4	0.518 ± 0.022	0.662 ± 0.035	0.322 ± 0.018	0.402 ± 0.039

Under the matched 57-dimensional input representation, the MLP achieved higher mean test scores than the classical comparator on all four tasks. Both model families generally performed better using the engineered descriptors than using the flattened raw signal values. These findings indicate that feature representation substantially influences the observed performance gap under the tested configurations.

For T1 and T2, exploratory paired bootstrapping over the nine held-out conditioning cells yielded performance differences (MLP minus classical) of 0.0059 in R^2 and 0.0085 in balanced accuracy, respectively, using the same 57-dimensional descriptors. The corresponding 95% bootstrap intervals were $[-0.1262, 0.0235]$ and $[-0.0038, 0.0225]$, both including zero.

The comparison does not establish superiority of either model family. It evaluates one MLP architecture against specific regularised classical models, and the training and validation-selection procedure differs from the original reference benchmark. In addition, the five training repetitions share the same test cells and do not constitute independent physical validation cohorts. The original reference results are therefore retained as descriptive benchmarks rather than replaced by this controlled comparison.

Structural invariants

Five structural invariants are tested automatically to verify release consistency and implemented bridge behaviour. First, at each OSF anchor voltage, the bridge's deterministic anchor-replica function (no perturbation, no noise) reproduces the empirical mean trajectory to within machine precision; the maximum absolute deviation observed is 1.87×10^{-12}.

Second, for fixed (voltage, SOC, seed), increasing the imposed SOH-conditioning magnitude produces monotonically increasing distance from the SOH=1.0 reference, confirming that the implemented conditioning operation behaves as specified.

Third, a 0.01 V change in voltage produces a smaller output change than a 0.10 V change (observed L2 changes ≈ 261 versus ≈ 1881), providing a numerical check of local voltage-response

continuity.

Fourth, the regime classifier returns the expected regime at boundary voltages and rejects out-of-range voltages. Fifth, anomaly flags are consistent with the released schema: every extrapolation-regime and synthetic anomaly sample carries `anomaly_flag` true, whereas every grounded clean sample carries it false.

In addition, the released artifacts were audited to confirm zero duplicate sample identifiers, zero duplicate full-signal hashes, no NaN or infinity entries in the signal channels, uniform length-100 arrays, and valid clean parents for all 600 synthetic anomaly samples.

Distributional comparison at grounded anchors

At the two grounded-regime OSF anchors (3.10 V and 3.34 V), 41 synthetic samples per anchor were generated at SOH=1.0 and compared with the corresponding real OSF samples. Per-channel two-sample Kolmogorov–Smirnov tests yielded $p > 0.05$ for all six channels at both anchors; these tests provide no evidence of a distributional difference at the tested sample size, but should not be interpreted as establishing distributional equivalence. Per-channel synthetic-to-real standard-deviation ratios lie in $[0.98, 1.02]$, indicating close agreement in marginal amplitude variability.

The mean absolute difference between synthetic and real cross-channel correlation matrices is 0.46 at 3.10 V and 0.35 at 3.34 V, indicating that cross-channel dependence is less closely reproduced than the marginal channel statistics. This residual is consistent with the bridge's independent-channel noise assumption and is documented as a known limitation in the Usage Notes.

SOH mechanism-knockout and direction controls

We performed matched ablations on 5,600 generation requests from 56 PulseBat conditioning cells, using a fixed cell-disjoint split of 39 training, 8 validation, and 9 test cells. All configurations shared the conditioning requests, random seeds, and downstream evaluation protocol. The released configuration (A0) served as the reference.

Table 5: SOH-conditioning ablations under the matched cell-disjoint regeneration protocol. T1 reports SOH-regression R^2, and T2 reports second-life balanced accuracy (BA). Values are mean ± standard deviation over five downstream model seeds. D2 reports the range of mean scores across five random-direction seeds (1001–1005). Model-seed variability does not represent between-cell uncertainty.

Configuration	T1 R^2	T2 BA
A0: Released configuration	0.6331 ± 0.0017	0.8222 ± 0.0046
S1: SOH-label shuffle	-1.0917 ± 0.0335	0.4911 ± 0.0053
D1: Reversed direction	0.6932 ± 0.0011	0.8261 ± 0.0047
D2: Random directions (range)	0.6424–0.7434	0.8226–0.8451
K1: No latent displacement	0.6457 ± 0.0017	0.8241 ± 0.0055
K2: No morphology blending	-1.0931 ± 0.0314	0.4916 ± 0.0099
K3: No amplitude jitter	0.6688 ± 0.0013	0.8279 ± 0.0017
K4: No spectral broadening	0.6395 ± 0.0018	0.8200 ± 0.0024
K5: All SOH operations disabled	-1.1011 ± 0.0299	0.4822 ± 0.0124

Shuffling SOH labels (S1) reduced performance substantially, confirming dependence on the imposed label–signal correspondence. Reversing the fitted direction (D1) or replacing it with random directions (D2) yielded higher mean T1 and T2 scores than the released configuration (A0). For T1, D1 achieved $R^2 = 0.6932$, while the five D2 directions yielded R^2 values ranging from 0.6424 to 0.7434, compared with 0.6331 for A0. These findings demonstrate that SOH-label recoverability does not require the particular fitted LDA direction used in the released bridge.

Among the mechanism knockouts, disabling morphology blending (K2) caused a substantial performance collapse, comparable to disabling all SOH-dependent operations (K5). In contrast, removing latent displacement (K1) or amplitude jitter (K3) increased mean performance on both tasks relative to the released configuration (A0). Removing spectral broadening (K4) increased T1 performance but slightly reduced T2 balanced accuracy. These results identify SOH-dependent morphology blending as the dominant conditioning pathway under the tested configuration, while demonstrating that the released parameters do not maximise downstream predictive performance. The knockout effects are not independent or additive because latent displacement influences the output through the decoding and blending pathway.

Exploratory paired cluster bootstrapping over the nine held-out conditioning cells yielded 95% confidence intervals excluding zero for the performance differences between A0 and both K2 and K5 on T1 and T2. These results support the importance of the SOH-dependent blending pathway, although the limited number of test cells restricts the precision and generalisability of the estimates.

Secondary cell-grouped cross-validation. Eight-fold GroupKFold evaluation across all 56 physical conditioning cells supported these findings. A0 achieved mean T1 $R^2 = 0.7839$ and T2 BA = 0.8767; K1 yielded 0.7863 and 0.8753, respectively. Performance decreased substantially for K2 (-0.0804, 0.4927) and K5 (-0.0829, 0.4925). These results reinforce the importance of morphology blending beyond the fixed nine-cell test partition.

Cone-orientation diagnostic. Aligning the decoder cone with the signed latent displacement instead of the released positive direction changed neighbour selection for 3,907 of 5,600 requests (69.8%) and increased T1 R^2 from 0.6331 to 0.6998. This demonstrates sensitivity to cone orientation but does not establish physical validity or justify modifying the frozen release.

Overall, these experiments demonstrate recoverability of synthetically imposed conditioning rather than an experimentally validated magnetic ageing mechanism.

Representation-dependence ablation

To test whether benchmark utility depends on the released quantum-reservoir encoder (QRec), we conducted a matched regeneration experiment. Four alternatives were evaluated: principal-component analysis (PCA), Gaussian random projection, radial-basis-function (RBF) kernel PCA, and a compact autoencoder. These cover linear, random, nonlinear-kernel, and learned representations.

Each representation produced a 171-dimensional embedding followed by the same four-dimensional LDA stage. All variants used identical 5,600 generation requests, conditioning records, per-sample seeds, bridge settings, decoder, and post-processing. The fixed 39/8/9 train/validation/test cell-disjoint split across 56 conditioning cells and the evaluation procedure were also identical; only the representation changed. Results are intended for within-ablation comparison and are not reproductions of the reference values in Table 3, which were evaluated directly on the frozen DOI-archived release.

For each representation, the reported value is the mean performance of the best candidate downstream model under the matched evaluation protocol (T1: ridge regression, RBF support-vector regression, or random forest; T2: logistic regression, ridge classifier, linear support-vector classifier, or random forest). The hyperparameter-sensitivity and generation-seed analyses instead use fixed ridge-regression and logistic-regression readouts.

The RBF kernel-PCA control used a fixed kernel parameter of $\gamma = 1/600$, while the autoencoder used a fixed $600{-}256{-}171{-}256{-}600$ reconstruction architecture. Neither alternative was tuned against downstream T1 or T2 performance.

For SOH regression (T1), the QRec, PCA, random-projection, RBF kernel-PCA, and autoencoder variants obtained mean R^2 values of 0.6331, 0.6717, 0.6975, 0.6686, and 0.6414, respectively. For second-life classification (T2), balanced accuracy was 0.8246, 0.8261, 0.8264, 0.8296, and 0.8195, respectively. Thus, no predictive advantage of the released QRec representation was observed under this matched ablation. T2 performance remained similar across the five representations, while PCA, random projection, and RBF kernel PCA produced higher T1 performance than QRec; the autoencoder produced a smaller T1 increase with slightly lower T2 performance.

These results indicate that the benchmark's decodable conditioning structure is not uniquely dependent on the quantum-reservoir representation. QRec is retained in MagBridge-Battery v1.0 because it is the representation used to generate the frozen DOI-archived dataset and is therefore required for exact reproduction of that release, not because it provides superior predictive performance. The matched ablation shows that simpler classical and learned representations are viable alternatives and, in some settings, achieve higher downstream performance. Accordingly, no quantum-performance benefit or quantum-advantage claim is made for QRec in this work.

Hyperparameter sensitivity

To assess whether benchmark behaviour depends strongly on the released bridge settings, we performed a one-factor-at-a-time sensitivity analysis over all 5,600 grounded generation requests. The tested values were $\gamma \in \{400, 800, 1200\}$, $k \in \{4, 8, 16\}$, $\tau \in \{25, 50, 100\}$, cone half-angle $\in \{45°, 75°, 90°\}$, and sensor-noise fraction $\in \{0.025, 0.05, 0.10\}$. All other bridge settings, conditioning records, seeds, and the same fixed 39/8/9 train/validation/test cell-disjoint regeneration split were held fixed. The released configuration is $\gamma = 800$, $k = 8$, $\tau = 50$, cone half-angle 75°, and sensor-noise fraction 0.05.

Table 6: One-factor-at-a-time sensitivity within the matched 5,600-request regeneration experiment. Ranges report the minimum and maximum mean performance observed across the three tested values of each parameter. T1 is SOH-regression R^2 and T2 is second-life balanced accuracy (BA).

Parameter	Tested values	T1 R^2 range	T2 BA range
γ	400 / 800 / 1200	0.6260–0.6406	0.8222–0.8247
k	4 / 8 / 16	0.6331–0.6394	0.8211–0.8258
τ	25 / 50 / 100	0.6329–0.6334	0.8214–0.8222
Cone half-angle	45° / 75° / 90°	0.6331–0.6495	0.8177–0.8222
Sensor-noise fraction	0.025 / 0.05 / 0.10	0.6309–0.6331	0.8222–0.8241

Across the tested ranges, the largest within-parameter change in T1 was 0.0164 in R^2 for the cone half-angle, while the largest T2 change was 0.0047 in balanced accuracy for k. The softmin temperature τ had negligible influence over the tested range. These results indicate that the benchmark conclusions are not strongly dependent on the particular released values of these parameters. The released settings should therefore be interpreted as fixed reproducibility defaults rather than as performance-optimised hyperparameters.

Generation-seed uncertainty

To quantify variability arising from stochastic generation, we independently regenerated the complete 5,600-sample grounded benchmark under five different master generation seeds. Each regeneration used the same 560 PulseBat-LFP conditioning records from 56 conditioning cells, ten variants per record, QRec/LDA representation, bridge hyperparameters, the same 39/8/9 train/validation/test split of those conditioning cells, and downstream evaluation protocol. Only

the random seed family used during synthetic generation was changed. For each regenerated dataset, T1 and T2 performance was averaged over the same five fixed model/subsampling seeds.

Across the five independently regenerated datasets, T1 SOH-regression performance was $R^2 = 0.6252 \pm 0.0145$, with individual-run means ranging from 0.6069 to 0.6440. T2 second-life balanced accuracy was 0.8203 ± 0.0157, with individual-run means ranging from 0.7985 to 0.8376. The reported standard deviations represent between-regeneration variability rather than within-model fitting variability.

These results show that stochastic generation introduces measurable variation in absolute benchmark scores, while all five complete regenerations retain comparable predictive structure for both reference tasks. Generation-seed uncertainty should therefore be considered when interpreting small performance differences on the benchmark. This analysis assesses robustness of the synthetic construction and does not establish physical validity of the imposed SOH-conditioned magnetic morphology.

Relationship to real measurements

The source datasets used here do not provide experimentally paired magnetic measurements and degradation labels for the same cells. Consequently, the SOH-conditioned synthetic signatures cannot be validated as measurements of real magnetic ageing trajectories. We therefore distinguish the aspects of the bridge that are directly constrained by measured OSF data from those introduced synthetically by the generation procedure.

Three observations characterize this relationship. First, at each of the five OSF voltage anchors, the deterministic anchor-replica function reproduces the corresponding empirical OSF mean trajectory to within machine precision (maximum absolute deviation 1.87×10^{-12}). This verifies the implementation of the measured anchor morphology but does not imply physical validation of subsequent SOH-conditioned transformations. Second, the grounded-regime base morphology is constructed by interpolation between measured OSF anchors, and the perturbation decoder retrieves from the OSF morphology bank. Thus, measured magnetic morphology directly constrains the generation procedure, although subsequent conditioning, scaling, spectral broadening, and noise remain synthetic operations. Third, at the two grounded anchors, per-channel Kolmogorov–Smirnov tests yielded $p > 0.05$ and synthetic-to-real standard-deviation ratios lie within $[0.98, 1.02]$. These results indicate close agreement in the tested marginal channel statistics at the available sample size, but do not establish distributional equivalence. Cross-channel correlation is reproduced less closely, as quantified above.

These checks do not establish that the SOH-conditioned morphology changes produced by the bridge correspond to those of a real cell at the same SOH. The benchmark results should therefore be interpreted as internal reference performance under the documented synthetic construction, rather than as estimates of performance on experimentally measured magnetic ageing data. The released provenance metadata and generation configuration are intended to enable direct comparison with paired magnetic–electrochemical measurements if such data become available.

Usage Notes

Loading the data

The released Parquet shards can be concatenated into a single table with any Parquet-aware data library. A minimal loader script, `load_example.py`, is provided in the code repository; it loads all shards, applies the `by_cell_primary` split, filters to the grounded regime, and plots an example signature following the released benchmark workflow. The signal channels and the `time_norm` column are stored as length-100 numeric arrays.

Recommended workflows

State-of-health regression (T1) and second-life classification (T2) should be performed on grounded-regime clean samples only; users should filter to `regime == "grounded"` and exclude anomalies before fitting these tasks, because extrapolation-regime samples carry `NaN` SOH by design. Anomaly detection (T3) uses a mixture of held-out grounded clean samples, synthetic anomalies, and extrapolation-regime samples. The three-class T3 formulation is intentional: a binary comparison between extrapolation-regime and grounded samples can largely reflect voltage-regime separation, whereas T3 additionally requires synthetic anomalies to be distinguished from clean grounded samples. Anomaly subtype classification (T4) should be performed on the synthetic sensor-anomaly subset using the four defined perturbation classes.

For reference benchmarking on the frozen, DOI-archived MagBridge-Battery v1.0 dataset, users should adopt the `by_cell_primary` split for all headline results. This split prevents conditioning-cell overlap and keeps synthetic anomaly samples in the same partition as their clean parents.

The `by_record_optimistic_baseline` split is provided only as a leakage-prone comparison and should not be used for headline benchmarking.

Controlled bridge-regeneration experiments reported in Technical Validation use a separate, fixed cell-disjoint split comprising 39 training, 8 validation, and 9 test conditioning cells. Results from these regeneration experiments are intended for within-study comparisons and should not be directly compared with the reference benchmark scores obtained using `by_cell_primary`.

What the benchmark tasks measure

T1 and T2 measure recovery of the SOH conditioning imposed by the synthetic bridge and the corresponding threshold-derived second-life label. Because the source datasets do not contain experimentally paired magnetic measurements and degradation trajectories for the same cells, these tasks do not measure detection of physical magnetic ageing, and their performance should not be interpreted as real-world SOH-estimation accuracy. T3 distinguishes clean grounded samples, synthetic sensor anomalies, and Regime-B samples, while T4 classifies the four synthetic sensor-anomaly subtypes. Neither task establishes the ability to detect or classify real battery or sensor faults.

Limitations users should be aware of

The dataset is synthetic and should be interpreted according to the assumptions of the bridge. The magnetic morphology bank is derived from measurements of a single physical cell in the OSF archive, so inter-cell variation in the underlying magnetic morphology is not represented directly. Moreover, the magnetic-morphology and electrochemical-conditioning sources are independent and are not chemistry matched: LFP in this release denotes the PulseBat conditioning population and should not be interpreted as the chemistry of the OSF magnetic source or of experimentally measured MagBridge signatures. Consequently, the imposed SOH-conditioned morphology changes have not been validated as physical magnetic ageing trajectories. SOH is used here as a scalar conditioning variable and does not uniquely encode degradation mechanism or ageing history; cells with similar SOH values may therefore represent different underlying degradation pathways.

Several construction-specific limitations should also be considered. The morphology bank is small, and the fitted latent perturbation direction is not physically unique or mechanistically causal; alternative-direction and representation ablations are reported in Technical Validation. The bridge applies per-channel noise independently, leaving residual differences between synthetic and real cross-channel correlation structure at the grounded anchors. The `time_norm` axis is a constant reference grid and carries no per-sample timing information. The source chan-

nels `B_s1C5` and `B_s2C6` are signed and should not be interpreted as strict physical magnitudes. Synthetic sensor anomalies are controlled benchmark perturbations rather than experimentally calibrated fault distributions for a specific magnetometer.

Because the SOH-conditioned morphology is generated through a synthetic conditioning procedure, models trained exclusively on MagBridge-Battery may learn characteristics of the bridge itself, including its interpolation, decoding, scaling, and noise structure, rather than physically meaningful magnetic degradation mechanisms. Predictive performance on the released benchmark should therefore be interpreted as evidence that a method can recover the imposed conditioning structure under the documented bridge assumptions, and not as evidence of real-world battery diagnostic accuracy.

The full-scale mechanism-knockout experiments further demonstrate that recovery of the imposed SOH labels depends predominantly on the synthetic morphology-decoding and blending pathway. Disabling SOH-dependent blending substantially reduced predictive performance, whereas removing latent displacement alone preserved substantial recoverability. Alternative perturbation directions also retained predictive information. These results show that benchmark performance does not uniquely validate the fitted LDA direction or establish a physically calibrated relationship between magnetic morphology and battery degradation. Users should therefore distinguish recovery of synthetically imposed conditioning from prediction of experimentally measured magnetic ageing.

The dataset is intended for reproducible benchmarking and method development, not for safety-critical battery decisions or direct estimation of performance on real magnetic ageing measurements. Every sample carries its generating provenance, including operating voltage, SOC, SOH where applicable, conditioning cell, seed, regime, anomaly fields, and bridge configuration hash, supporting reproducible analysis and future comparison with paired measurements.

Data Availability

The MagBridge-Battery dataset is available from Zenodo [20] at `https://doi.org/10.5281/zenodo.20260147`.

Code Availability

The bridge implementation, generation scripts, benchmark suite, and unit tests are openly available on GitHub at `https://github.com/SakthiGs/MagBridge-Battery` under the Apache-2.0 licence. The repository includes the code used to generate MagBridge-Battery, the OSF-derived artifacts required to run the bridge, a minimal loader script (`load_example.py`), and benchmark scripts for reproducing the reference values reported in Table 3.

The controlled bridge-regeneration experiments reported in Technical Validation use separate analysis scripts and a fixed experimental protocol. These experiments include SOH-label shuffling, alternative-direction controls, mechanism knockouts, grouped cross-validation, and cone-orientation diagnostics. The experimental configuration, input provenance, random seeds, and evaluation procedures are documented to distinguish these analyses from the reference benchmarks performed on the frozen dataset release.

The reproducibility archive contains the experimental protocols, analysis scripts, software environment records, input checksums, random seeds, per-configuration predictions, evaluation metrics, and bootstrap outputs for the additional validation experiments. These materials support independent inspection of the reported results and distinguish the controlled regeneration experiments from the reference benchmarks performed on the frozen dataset release.

Supporting analysis materials for the controlled validation experiments, including configurations, random seeds, per-configuration predictions, evaluation metrics, and bootstrap outputs,

are publicly available at https://github.com/SakthiGs/MagBridge-Battery/releases/tag/validation-evidence-2026-11-11.

The software was tested with Python 3.10–3.12 and uses standard scientific-Python libraries including NumPy, pandas, scikit-learn, and PyArrow. The reservoir encoder uses PennyLane, and the neural reference baselines use PyTorch. The released configuration file records the bridge hyperparameters, and the generated dataset records the corresponding bridge configuration hash.

The default test discovery collects 196 test items. Two tests marked as slow are deselected by the repository's default pytest configuration; among the remaining 194 selected tests, a clean public-repository run reports 163 passed and 31 skipped. Tests requiring unavailable raw upstream files are skipped automatically.

References

[1] Severson, K. A. *et al.* Data-driven prediction of battery cycle life before capacity degradation. *Nat. Energy* **4**, 383–391 (2019). https://doi.org/10.1038/s41560-019-0356-8

[2] Tao, S. *et al.* Generative learning assisted state-of-health estimation for sustainable battery recycling with random retirement conditions. *Nat. Commun.* **15**, 10154 (2024). https://doi.org/10.1038/s41467-024-54454-0

[3] Tao, S. & Ma, G. Pulse Voltage Response Generation: TBSI-Lijing Battery Dataset (v3). Zenodo (2024). https://doi.org/10.5281/zenodo.13360631

[4] Braco, E., San Martín, I., Sanchis, P., Ursúa, A. & Stroe, D.-I. State of health estimation of second-life lithium-ion batteries under real profile operation. *Appl. Energy* **326**, 119992 (2022). https://doi.org/10.1016/j.apenergy.2022.119992

[5] Camboim, M. M. et al. State of health estimation of second-life batteries through electrochemical impedance spectroscopy and dimensionality reduction. *J. Energy Storage* **78**, 110063 (2024). https://doi.org/10.1016/j.est.2023.110063

[6] Wang, S., Hu, B., Zhang, Q., Dang, Q. & Fernandez, C. Critical review on improved hybrid deep learning modeling strategies for online battery state of charge estimation in large-scale energy storage systems. *J. Power Sources* **673**, 239714 (2026). https://doi.org/10.1016/j.jpowsour.2026.239714

[7] Wang, S., Wei, J., Zhang, L., Li, H., Fernandez, C. & Blaabjerg, F. Improved harmonic loss–History gated unit recycling for online state of charge and state of energy co-estimation of lithium-ion batteries for large-scale energy storage stations. *Energy* **340**, 139225 (2025). https://doi.org/10.1016/j.energy.2025.139225

[8] Wang, S., Ma, C., Gao, H., Deng, D., Fernandez, C. & Blaabjerg, F. Improved hyperparameter Bayesian optimization-bidirectional long short-term memory optimization for high-precision battery state of charge estimation. *Energy* **328**, 136598 (2025). https://doi.org/10.1016/j.energy.2025.136598

[9] Ilott, A. J., Mohammadi, M., Schauerman, C. M., Ganter, M. J. & Jerschow, A. Rechargeable lithium-ion cell state of charge and defect detection by in-situ inside-out magnetic resonance imaging. *Nat. Commun.* **9**, 1776 (2018). https://doi.org/10.1038/s41467-018-04192-x

[10] Romanenko, K., Kuchel, P. W. & Jerschow, A. Accurate visualization of operating commercial batteries using specialized magnetic resonance imaging with magnetic field sensing. *Chem. Mater.* **32**, 2107–2113 (2020). https://doi.org/10.1021/acs.chemmater.9b05238

[11] Hu, Y. *et al.* Sensitive magnetometry reveals inhomogeneities in charge storage and weak transient internal currents in Li-ion cells. *Proc. Natl. Acad. Sci. USA* **117**, 10667–10672 (2020). https://doi.org/10.1073/pnas.1917172117

[12] Ilott, A. J. & Jerschow, A. Super-resolution surface microscopy of conductors using magnetic resonance. *Sci. Rep.* **7**, 5425 (2017). https://doi.org/10.1038/s41598-017-05878-w

[13] Pollok, S. *et al.* Magnetic microscopy for operando imaging of battery dynamics. *Nat. Commun.* **16**, 8303 (2025). https://doi.org/10.1038/s41467-025-63409-y

[14] Evans, W. *et al.* Quantum magnetic imaging of current density in lithium-ion batteries. Preprint at https://arxiv.org/abs/2512.01125 (2025).

[15] Fraunhofer IFAM. Quantum technology and AI: the key to safe second-life applications for lithium-ion batteries. Press release (30 January 2025). https://www.ifam.fraunhofer.de/en/Press_Releases/quantum-technology-and-AI-second-life-applications-lithium-ion-batteries.html

[16] Mohammadi, M. & Jerschow, A. Battery Magnetometry Data 2019–2020. Open Science Framework (2020). https://doi.org/10.17605/OSF.IO/CW8ZV

[17] Fujii, K. & Nakajima, K. Harnessing disordered-ensemble quantum dynamics for machine learning. *Phys. Rev. Appl.* **8**, 024030 (2017). https://doi.org/10.1103/PhysRevApplied.8.024030

[18] da Silva, J. C., Saxena, A., Balaban, E. & Goebel, K. A knowledge-based system approach for sensor fault modeling, detection and mitigation. *Expert Syst. Appl.* **39**, 10977–10989 (2012). https://doi.org/10.1016/j.eswa.2012.03.026

[19] Janošek, M., Dressler, M., Petrucha, V. & Chirtsov, A. Magnetic calibration system with interference compensation. *IEEE Trans. Magn.* **55**, 1–4 (2019). https://doi.org/10.1109/TMAG.2018.2874169

[20] Gunasekar, S. P. & Rangarajan, P. K. MagBridge-Battery: A synthetic magnetic-signature dataset with electrochemical degradation labels for lithium-ion batteries (v1.0). Zenodo (2026). https://doi.org/10.5281/zenodo.20260147

Author Contributions

S.P.G. designed and implemented the bridge, generated the dataset, carried out the technical validation and benchmarking, and drafted the manuscript. P.K.R. supervised the project and contributed to revising the manuscript. Both authors reviewed and approved the final manuscript.

Competing Interests

The authors declare no competing interests.

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